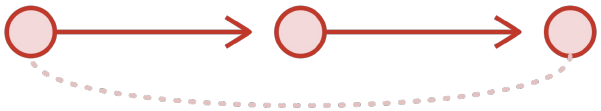
The Modern Bank SOC Stack

At Meridian Trust Bank, this stack works together as one connected pipeline rather than four separate tools.



1 Detect



Detect

SIEM and EDR generate alerts from correlated log and endpoint data. This is where nearly every investigation begins.

In practice

- SIEM correlates firewall, DNS, and authentication logs to flag activity that breaks from baseline.
- EDR watches endpoint telemetry — process creation, file changes, registry edits — for suspicious behavior.
- Alerts are auto-scored by severity so analysts triage the highest-risk items first.
- A single raw event rarely becomes an alert on its own; it's the correlation across sources that does.

2 Investigate

Investigate

Analysts use SPL and KQL queries, often assisted by Copilot, to dig into the alert's full context. This is the skill built throughout today's session.

In practice

- SPL and KQL pull the full event timeline surrounding an alert, not just the triggering event.
- Copilot can suggest starting queries and surface related indicators, but the analyst validates every result.
- Findings get mapped to MITRE ATT&CK techniques to scope what the attacker actually did.
- Good investigation answers three questions: what happened, what else was touched, and is it still active.

3 Respond

Respond

SOAR executes automated actions for well understood cases, while analysts handle escalation and incident response for the rest. This closes the loop back to the Week 7 content on IR and automation.

In practice

- SOAR playbooks auto-isolate hosts, disable compromised accounts, and block known-bad indicators.
- Anything outside the playbook's confidence threshold escalates to an analyst for manual handling.
- Escalated incidents follow the IR process: containment, eradication, recovery, and lessons learned.
- Every automated and manual action is logged, feeding the post-incident review and future playbook tuning.

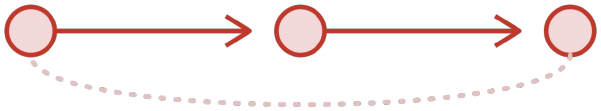
ONE CONNECTED PIPELINE



Four tools. One pipeline. The handoffs between Detect, Investigate, and Respond are what make the SOC work.

Case Study: A Full Stack Response

A recent incident at Meridian Trust Bank shows all four layers of the stack working together.



1 Detection

Detection

Sentinel correlated an EDR alert about unusual process activity with a sign-in log anomaly, generating a combined, higher confidence alert. Neither source alone would have carried the same weight.

REAL-WORLD PARALLEL *Intesa Sanpaolo, Microsoft Sentinel + Copilot for Security*

- Intesa Sanpaolo runs Microsoft Sentinel as its global SIEM, correlating signals across hundreds of branches on every continent.
- Real-time threat intelligence feeds are embedded directly into Sentinel to catch “unknown unknowns” faster.
- “Visibility is our most effective weapon against cyber unknown unknowns.” — Matteo Feraboli, Group Sr. Director, Cyber Security & Anti-Fraud, Intesa Sanpaolo

2 Investigation

Investigation

An analyst used a KQL query, refined with a Copilot follow-up prompt, to confirm the scope of affected systems within minutes. This combined today's query skills with last week's prompting skills directly.

REAL-WORLD PARALLEL *Intesa Sanpaolo, Microsoft Sentinel + Copilot for Security*

- Junior analysts at Intesa get productive fast using Copilot for Security prompts to generate KQL scripts instead of writing them from scratch.
- Senior analysts use it too — the script analyzer doubles as on-the-job training by surfacing the malicious patterns behind each query.
- Microsoft reports customers using Security Copilot see a 30% reduction in mean time to respond (MTTR).

3 Response

Response

A SOAR playbook isolated the affected endpoint automatically while the analyst began the formal incident response process from Week 7. Each layer of the stack did exactly what it was built for.

REAL-WORLD PARALLEL *Intesa Sanpaolo, Microsoft Sentinel + Copilot for Security*

- Intesa's team uses Copilot's reporting feature to turn incident resolution and KPIs into executive briefings in minutes, not hours.
- That frees analysts to refocus on active defense instead of manual write-ups.
- Faster reporting keeps executives informed and ready to back the next security initiative.

THE OUTCOME

Every layer of the stack did exactly what it was built for.

Detection

Investigation

Response

Detection flagged it. Investigation scoped it. Response contained it — no layer carried the incident alone.